

Different information horizons shape wind-power event predictability

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Abstract

Wind-power events are often reduced to a single threshold label, although the event state and the time at which a process first arrives carry different information. We study these forecast objects on a common Pizhou wind-power target at issue leads from 15 min to 12 h. We separate state risk from first-passage timing and use matched external information arms to identify which source serves each target. Event-state risk and first-passage timing show different information horizons, and the power-history timing pattern is retained at two site datasets, including an independent La Haute Borne site. In a matched three-arm Suining decomposition, the weather-only arm retains approximately 10% four-class skill relative to the frequency reference across the six leads, while adding power history to the weather arm improves downward timing RPS by 10.17% at 15 min and 6.32% at 60 min, with corresponding median-MAE improvements of 11.06% and 6.75%. The archived weather trajectory provides a retrospective decomposition, while the strict previous_day1 package is fixed 24 h before valid time within the NP028 Suining support window. Under this strict package, weather adds 12.11% to the 12-h state-risk score relative to power history, whereas power history adds 7.83% to 15-

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min timing RPS and 7.40% to median-MAE relative to weather. Routing the joint state head and the power-history timing head improves timing RPS by 8.70% and 8.65% at 240 and 480 min, respectively, with absolute conditional RPS changing from 0.183 to 0.167 and from 0.182 to 0.167. The route preserves state probabilities exactly. A coherent 17-bin check shows that the route exceeds a direct joint model at 240–720 min under the strict clock, while the archived comparison remains mixed. These findings support a target-aligned information rule for the tested sites and matched issue-time contracts.

Keywords: wind power forecasting, event predictability, first-passage timing, numerical weather prediction, power history, target-aligned information

1. Introduction

Wind-power forecasting has progressed from single-valued trajectory prediction toward probabilistic products that quantify uncertainty and support operational decisions^{1–5}. This progression is especially important for rapid power changes. Ramp events are defined through combinations of magnitude, duration, direction, and temporal window, so the same trajectory can receive different event labels under different operational rulers^{6–9}. Average MAE or RMSE therefore hides the part of the forecast that determines reserve, balancing, and alert actions.

A process forecast contains several distinct objects. Occurrence risk asks whether a threshold excursion will appear; first-passage timing asks when the excursion will arrive; amplitude and path order describe how the excursion develops. Existing ramp studies have improved event detection with NWP ensembles, meteorological pattern analysis, mesoscale models, and direct wind observations^{7,10–13}. Other work has developed event-oriented metrics and systematic forecast validation frameworks^{8,14,15}. These studies establish the operational importance of process forecasts, while their main comparisons still tend to aggregate the process into one event score or one forecast trajectory.

Probabilistic forecasting provides the tools to separate these objects. Scenario generation preserves dependence across future time points^{16–18}; spatially distributed information improves conditional wind-power distributions¹⁹; and ensemble calibration, proper scores, and sharpness diagnostics determine whether probability gains are usable^{20–26}. Recent NWP post-processing and deep probabilistic models further show that forecast skill depends on the timing of weather information, the target horizon, and the representation used by the model^{27–34}. The

unresolved question is how these information conditions map onto different process objects within the same target window.

Weather conditions add a second source of conditional variation. Synoptic regimes, mesoscale ensembles, hub-height NWP errors, and local observations influence when a power process becomes predictable^{10,11,15,35–38}. Yet a weather product can contribute broad state information without resolving the local phase of an internal reversal. A power history can carry local phase information at short lead while losing value at longer lead. The relevant scientific comparison is therefore target-conditional: which information arm reduces the loss for which process object and at which issue lead?

Wind-power forecasting has moved from single-valued trajectories toward probabilistic products that represent uncertainty for operational decisions^{1–3,5}. Ramp studies have complemented this trajectory view with NWP-ensemble forecasts, mesoscale ensembles, weather-regime conditioning, meteorological-pattern analysis and local wind observations^{7,10–13}. Scenario methods and joint time-distribution models further represent dependence across future times when converting wind-speed forecasts into ramp probabilities^{16,17}. Event-oriented metrics evaluate timing, duration, amplitude or region-of-interest performance, while proper scores and calibration methods make probabilistic improvements interpretable^{8,15,21,22,24–26}. Together, these studies establish the value of weather information, temporal dependence, event-aware scoring and calibrated uncertainty. Their primary comparisons usually centre on an aggregate ramp event, a single forecast product or one model-input configuration; the evidence therefore motivates a matched comparison of which process object benefits from which issue-time information at which lead, without requiring a first-study claim.

We formulate that comparison as a target-object $\times$ input $\times$ lead decomposition. The target objects are event-state risk, first-passage timing, amplitude/range and path order; the input arms are issue-time power history, issued or archived weather, and their matched joint combination; and the lead axis spans the same target window under a fixed maturity and release-time contract. For each object and lead, we compare paired proper-score losses under the same windows and model budget, then test whether the resulting information assignment can be expressed as a target-aligned route that retains the joint state head while assigning the timing head to the information arm with the relevant timing skill. This design extends the published GNSS-guided short-horizon forecast contribution³⁹ and connects it to recent work on NWP timing alignment, hub-height ramp evaluation and probabilistic wind-power forecasting^{27–31,35}; it tests a process-level information rule under explicit site and issue-time protocols rather than introducing a new network

architecture.

Our published GNSS study established that an additional atmospheric observation can improve short-horizon wind-power forecasting³⁹. The present paper develops the next scientific step using established event labels and probabilistic forecast tools: it separates event-state risk from first-passage timing, measures information value on matched windows, and converts the decomposition into a target-aligned product. This distinction is related to the broader forecast-to-decision literature, where forecast value depends on the user’s loss function and operational action^{5,40–44}.

Here we study the conditional predictability of wind-power event structures (Fig. 1). We ask three questions. First, which process objects retain useful information as the issue lead increases? Second, do archived weather information and power history contribute to state risk and first-passage timing in the same way? Third, can this information decomposition be converted into a target-aligned timing product using issue-time inputs?

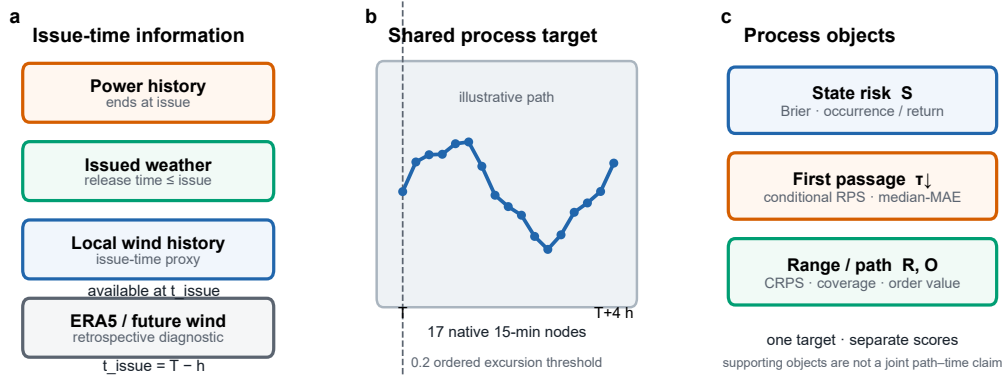


Figure 1: Process objects and information boundary.

The study makes three contributions. First, it identifies distinct information horizons for event-state risk and first-passage timing, with a power-history process-object replication at La Haute Borne and Suining. Second, it quantifies target-conditional information value with matched power, archived weather, and joint arms, then repeats the state-risk comparison under a strict 24-h previous_{day1} clock. Third, it converts the decomposition into a target-aligned route that assigns the joint state head and the power-history timing head to their respective process products. The result is a measurable information-to-target rule supported under the tested wind-power protocols.

2. Data and methods

2.1. General process-object forecast functional

Let T denote the beginning of a four-hour target and $t_{\text{issue}} = T - h$ the issue time at lead h . Let q index a forecast object and a index an issue-time information arm. The study can be written as a family of conditional forecast functionals:

$$\mathcal{Y}_{q,T} = \Phi_q(Y_T), \quad \hat{F}_{q,h}^a = f_q(X_{t_{\text{issue}}}^a, h), \quad \mathcal{L}_{q,h}^a = \mathbb{E} \left[S_q \left(\hat{F}_{q,h}^a, \mathcal{Y}_{q,T} \right) \right],$$

where Φ_q extracts the process object, S_q is its proper score, and $\mathcal{L}_{q,h}^a$ is the expected forecast loss under the information arm a . The target-conditioned value of adding information B to reference information A is

$$\text{IV}_q(A \rightarrow B; h) = 100 \frac{\mathcal{L}_{q,h}^A - \mathcal{L}_{q,h}^B}{\mathcal{L}_{q,h}^A}.$$

For a multiclass state vector (Z_T) evaluated with the Brier score, the corresponding population identity is

$$\mathcal{L}_{\text{Brier}}(A) - \mathcal{L}_{\text{Brier}}(A, B) = \mathbb{E} \left[\|P(Z_T | X_A, X_B) - P(Z_T | X_A)\|_2^2 \right].$$

The Brier identity is a population identity for Bayes-optimal conditional probabilities. In this study, IV_q is instead a model-relative empirical loss reduction measured on paired test windows under fixed feature arms, tree budgets and issue-time contracts; it is an operational information-value statistic rather than a causal estimand or a population mutual-information measure. Added information has value when it changes the conditional state distribution in a way the forecast can resolve. Conditional RPS applies the same target-specific comparison to the cumulative first-passage distribution.

The main objects are $q = S$ for event-state risk and $q = \tau_{\downarrow}$ for conditional first passage. The matched decomposition estimates IV_q on the same windows and model budget; the routing test evaluates

$$\hat{\mathbf{F}}_h^{\text{route}} = (f_S(X_J, h), f_{\tau_{\downarrow}}(X_P, h)) \quad \text{against} \quad \hat{\mathbf{F}}_h^J = (f_S(X_J, h), f_{\tau_{\downarrow}}(X_J, h)).$$

This formulation connects the three parts of the study: process objects define the target, information arms define the available evidence, and issue lead defines the prediction contract.

2.2. Site and information boundary

The main information-arm study uses the Pizhou wind-power series and a 15-min native target (Table 1). Pizhou power is normalized by the provider-supplied 87.45-MW farm capacity; the 0.2 excursion threshold is applied on this scale. Suining and the external process sites use the same normalized excursion rule with their site-specific training-period q99.5 scales. Thus, the process definition is shared while the physical scaling remains site-specific. The primary spatial reference is the administrative coordinate 34.3403°N, 118.0068°E. The user-provided onsite wind reference height is 140 m. GFS 100 m, JMA 10 m, ERA5 100 m, and onsite wind measurements are retained as separate products. No height conversion is applied. External process-object validation uses 15-min SCADA from La Haute Borne and Suining. The Suining external information decomposition uses an archived historical GFS forecast trajectory with 100-m wind, 1000-hPa temperature and surface pressure interpolated to the target nodes; its product timestamps and response hashes are retained in the Supplementary material. A separate strict-clock validation uses GFS previous_day1 100-m wind, surface pressure and 2-m temperature, each fixed 24 h before valid time. Matched power-only, weather-only and joint arms share the same windows and model budget.

ERA5 is used retrospectively to describe atmospheric context and to construct same-height diagnostic references⁴⁵. Future onsite wind is used only in explicitly labelled retrospective diagnostics or as a matured historical training target. It is never passed to an operating prediction at the target issue time. This boundary separates information that can be used at issue time from information that can only be used to diagnose an information gap after the event.

Table 1. Data and protocol

Item	Value
Main site	Pizhou administrative reference: 34.3403°N, 118.0068°E; onsite wind reference 140 m
Target	17 native 15-min power nodes over [T, T+4 h]; excursion threshold 0.2 pu
Issue leads	15, 60, 120, 240, 480, 720 min
Main test cohort	6,496 target windows

Item	Value
External decomposition	Suining; power-only, weather-only, and joint arms; 17 nodes; 60/20/20 chronological split with four-hour purge; 6,996 test windows per lead
Strict weather clock	GFS previous_day1 fields fixed 24 h before valid time; request January–August 2024; complete weather support from 16 February; 3,772 test windows per lead
Target-aligned route	Joint state head plus power-history timing head; frozen NP025 predictions, no refit

2.3. Process target and evaluation objects

Each target starts at T and contains the current node plus 16 future 15-min intervals:

$$Y_T = (P_T, P_{T+15\text{ min}}, \dots, P_{T+4\text{ h}}),$$

with 17 native nodes and 16 intervals. An issue lead h means that the forecast inputs stop at $t_{\text{issue}} = T - h$. The upward and downward excursions are defined from running extrema within Y_T : $U_T = \max_j (P_{T+j} - \min_{i \leq j} P_{T+i})$ and $D_T = \max_j (\max_{i \leq j} P_{T+i} - P_{T+j})$. The threshold is 0.2 on the Pizhou provider-capacity scale and 0.2 on each external site’s training-period q99.5 scale. The four process classes are neither, upward, downward, or both directions. The target values retain the recorded power support, including values slightly outside the nominal $[0, 1]$ range; no clipping is applied in the primary analysis.

The main forecast objects are the state risk S_T and the downward first-passage time $\tau_{T,\downarrow}$. The first-passage implementation checks the 16 future nodes after T ; bin 0 is 15 min, bin 15 is 240 min, and bin 16 denotes no crossing. For direction d , let $D_{T,d}$ indicate that the corresponding excursion occurs and let $\tau_{T,d}$ be its first threshold crossing within the four-hour window. The timing head estimates a conditional distribution over $\tau_{T,d}$ on windows with $D_{T,d} = 1$. In NP009, the context arm uses endpoint/context covariates, the ordered arm uses the ordered NWP interior trajectory, and the sorted arm independently sorts the same interior values while retaining endpoint/context features; the timing model is fitted on matured positive windows with the validation-selected pooled λ . NP009 combines

this timing distribution with the frozen NP008 full rolling occurrence marginals; NP025 and NP028 hold occurrence at the mature training frequency for their external timing arms. The conditional timing denominators are 2,653 downward events per lead in NP025 and 1,328 in NP028, within full test cohorts of 6,996 and 3,772 windows. Occurrence probabilities are scored with multiclass, upward, downward, and return-event Brier scores. First-passage timing is scored with conditional ranked probability scores and median arrival-time error. As a joint accounting diagnostic, the conditional timing distribution is multiplied by the downward occurrence probability and given a no-crossing bin; the resulting 17-bin RPS is reported in the Supplementary material. The multiclass Brier score is the sum of squared probability errors over the four state classes; conditional timing RPS is the mean squared CDF error over the 15 crossing thresholds; the frequency reference uses the training-cohort class distribution. Range and path are supporting marginal objects scored with CRPS, interval coverage, interval width and downstream event value. These scores keep state risk, timing, amplitude and path order as distinct forecast objects.

2.4. Chronological protocol

The main chronological target uses six issue leads: 15, 60, 120, 240, 480, and 720 min (Table 1). Training, validation, and test windows are shared across lead comparisons whenever the information arm is complete. Rolling probability updates use only labels that mature at least T+4 h+1 min before the update. Calendar-block bootstrap intervals use the same target blocks for the compared methods. The test period contains 6,496 target windows.

2.5. Information arms and target alignment

The information arms include power history, issued GFS/JMA weather, onsite wind history, retrospective ERA5 context, predicted local-wind paths, predicted wind ranges, and combinations of frozen rolling probabilities. The historical-forecast package used in NP024/NP025 is kept distinct from the strict previous_day1 package in NP028/NP029. NP025 indexes each archived forecast value by its target-valid timestamp and rejects rows with missing required fields; because the historical archive does not expose a model release timestamp, this arm is interpreted as a retrospective information decomposition rather than a strict real-time release test. For the NP025 external three-arm decomposition, the test cohort contains 6,996 windows per lead; the power arm contains 16 issue-time power lags and target calendar terms, the weather arm contains target calendar terms and the four-variable historical-forecast trajectory, and the joint arm concatenates both.

The strict NP028 package uses 3,772 test windows per lead, complete weather support from 2024-02-16, 100-m u/v wind, surface pressure and 2-m temperature, with each value fixed 24 h before valid time. NP031 repeats the event and timing fits using only the four known target-calendar variables. The target-aligned route is a frozen transformation of NP025 predictions: it copies the joint state head and the power-history conditional timing head, adding a direct target-to-head mapping for the process product. The route mapping is fixed by the decomposition protocol before test-score interpretation; no test lead or month is selected. The NP013 and NP014 paths are generated from past matured wind labels and issue-time inputs. The empirical leaf distributions follow the adaptive-neighborhood interpretation of random forests⁴⁶. The NP014 rank assignment is an aggregate range adaptation motivated by ensemble copula coupling¹⁸. The NP016–NP020 calibration and fusion experiments reuse frozen rolling probabilities and do not refit their source models. All information-value statements below are model-relative comparisons under these fixed feature contracts and budgets; they identify the arm associated with a lower loss for a process target, rather than a causal effect of an isolated atmospheric variable.

The main target-to-information mapping is fixed before interpretation: process-state risk uses the joint power-weather arm; first-passage timing uses the target-aligned power-history timing head; wind-range uncertainty uses the direct range arm; and decision sensitivity uses frozen return-event probabilities. For a candidate B and reference A , information value is reported as $100[L_q(A) - L_q(B)]/L_q(A)$ for the same task, windows and proper score. Each object is therefore interpreted within the process target for which it was designed.

The complete reproducibility protocol, including cohort sizes, maturity, model budgets, score definitions, absolute scores, the no-crossing-inclusive first-passage accounting, calendar-only control, weather-variable ablation and block resampling, is provided in the current paper-specific Supplementary material. All information arms use the same tree count, depth and leaf-size budget; feature dimensionality remains part of the model-relative comparison and is exposed by NP031/NP034 rather than treated as a causal control.

2.6. Statistical estimation and reproducibility

Rolling event models use the preceding 56 days and update at UTC Monday 00:00. A label enters a fit only after the complete four-hour target has matured. The principal ExtraTrees configuration uses 150 trees, depth 16, minimum leaf size 32, all available features at each split, and seed 41; the 450-tree model is retained only where it is explicitly labelled as a capacity control. The logistic calibration

arm uses an L2 penalty with $\lambda=0.001$. Relative score changes are paired on the same target windows. The main uncertainty intervals use 2,000 calendar-block resamples with seven-day blocks, with three- and 14-day blocks as sensitivity checks. These intervals condition on the fitted models and do not include refitting uncertainty. The full event test cohort has 6,496 windows; wind-complete diagnostics use the stated 6,041 or 6,059 subsets. Lead-wise and arm-wise contrasts are pre-specified descriptive endpoints; their paired intervals are reported without family-wise multiplicity adjustment. The decision-value sensitivity uses only frozen return-event probabilities: alert thresholds are selected on validation data for fixed false-negative/false-positive cost ratios of 2, 5, and 10, then applied once to the test cohort. NP031 uses the same train/test masks and model budgets with only the four known target-calendar variables, providing a calendar-only information control. These costs are study-defined sensitivities rather than a market invoice.

3. Results

3.1. Process-state risk and first-passage timing have different useful horizons

The first result is that one four-hour event target contains process objects with different information horizons in the Pizhou NP005/NP008/NP009 protocol (Fig. 2; Table 2; Fig. 3). Across the six issue leads, return-event risk and downward first-passage timing follow different skill trajectories. The NP005 power-plus-weather joint classifier reduces 12-h return-event Brier loss by 5.37% relative to its training-frequency reference. After the occurrence probability is held at its mature training frequency, the 15-min downward first-passage median error decreases from 54.22 to 49.35 min and conditional RPS loss decreases by 8.14%. The timing increment becomes smaller as the issue lead increases, whereas broad state-risk information remains available at longer leads. The two objects therefore provide complementary forecast outputs: a state-risk vector over all windows and a conditional arrival distribution on downward-event windows.

Table 2. Pizhou NP005/NP008/NP009 process skill across issue lead

Process object	Lead	Relative loss reduction
Four-class state: full vs weather	15 min	10.27% (8.18, 12.81)
Four-class state: full vs weather	720 min	-0.25% (-0.72, 0.22)
Return-event risk	720 min	5.37% (0.20, 11.80)
Downward first-passage RPS	15 min	8.14% (5.43, 10.85)

Process object	Lead	Relative loss reduction
Downward first-passage median MAE	15 min	8.98% (5.67, 11.94)

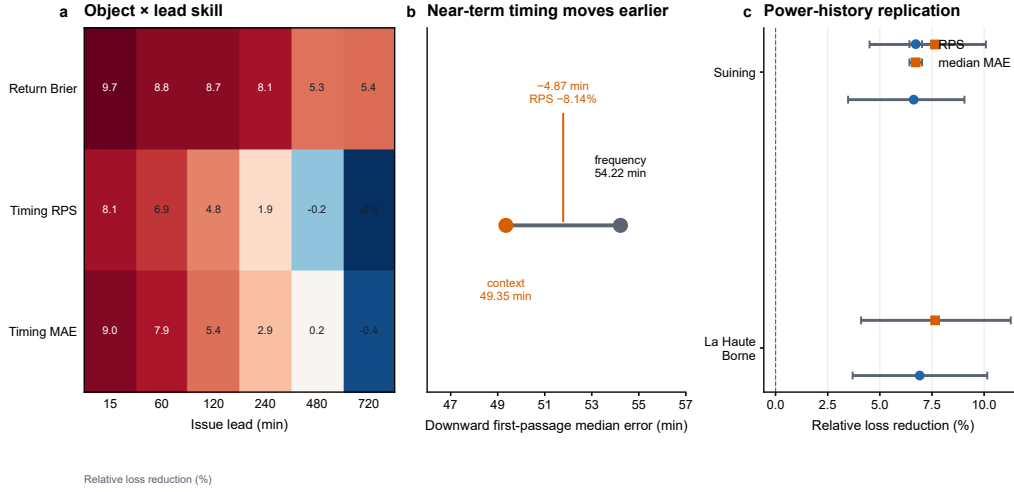


Figure 2: Process-state and first-passage skill retain different issue-time horizons. Points show frozen test scores; vertical ranges show the paired seven-day block bootstrap intervals (2,000 re-samples).

The near-term timing pattern is retained in a no-refit calendar audit and in two site datasets: La Haute Borne and Suining improve 15-min downward timing RPS by 6.91% (3.69–10.15%) and 6.63% (3.48–9.06%), and median arrival-time MAE by 7.65% (4.10–11.28%) and 7.64% (4.51–10.09%), respectively. La Haute Borne is an independent external farm; Suining is the site used for the matched information decomposition. These results define two complementary forecast objects for the information decomposition: event-state risk and first-passage timing.

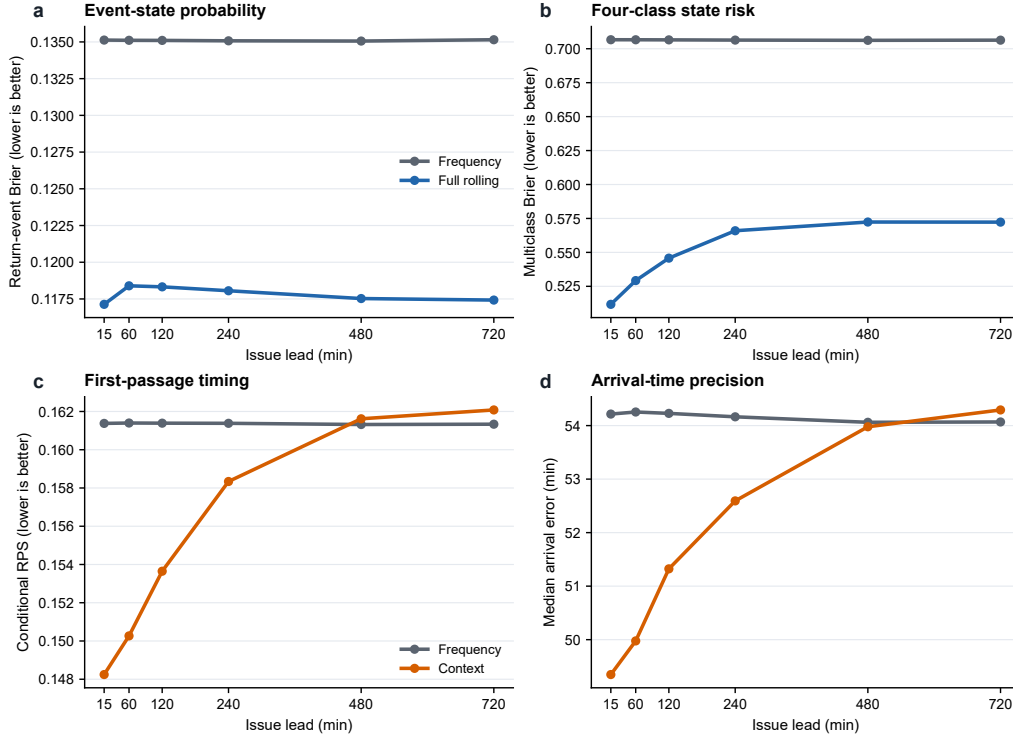


Figure 3: Pizhou core state-risk and first-passage skill across issue lead (NP005/NP008/NP009; 6,496 target windows). Points show absolute scores and intervals show paired seven-day block bootstrap uncertainty.

3.2. Issued weather and power history contribute to different process targets

The matched Suining decomposition assigns information value by process target (Fig. 4). The weather-only arm retains 9.92–10.06% four-class Brier skill relative to the frequency reference across the six leads. Its relative contribution shifts with lead, moving from the short-lead power-dominated regime toward a positive weather increment at longer leads. The joint arm improves the multiclass score over power history at every lead and reaches 9.78% (4.61–15.15%) at 12 h. Adding power history to the weather arm improves downward timing RPS by 10.17% (7.06–12.99%) at 15 min and 6.32% (3.23–9.31%) at 60 min; the corresponding median-MAE improvements are 11.06% (8.50–13.26%) and 6.75% (3.49–9.31%). The calendar-only control remains below the frequency reference for both state and timing objects, showing that the measured increments arise from the power and weather variables beyond shared calendar terms. A three-seed refit sensitivity preserves the long-lead state increment: joint-versus-power multiclass

Brier gains average 9.56% (SD 0.48%) for NP025 and 12.61% (SD 0.57%) for NP028 at 12 h. The variable ablation in Fig. 5 further shows that the state signal is carried primarily by the issued wind vector: wind-only gains remain positive at 15, 240 and 720 min, whereas thermodynamic-only inputs remain below the frequency reference.

The same target-aligned principle appears in the supporting range pathway. Direct range prediction improves wind-range CRPS by 18.57% (15 min) and 16.61% (12 h) on 6,059 complete-wind windows, while the path-order assignment adds a 1.12% return-event reduction at 12 h. The corresponding intervals, absolute CRPS values and matched comparator are listed in Supplementary Table S15. These results separate amplitude and order information from the state and timing products. A threshold audit on the same Suining cohorts at 0.15 and 0.25 preserves positive weather-only state point gains at every tested package–lead combination (NP025: 9.70–9.95%; NP028: 4.02–7.45%), while power history adds 4.52–10.75% to 15-min timing RPS relative to weather. The matched decomposition therefore identifies the weather arm with lower state-risk loss and the power-history arm with lower near-term first-passage loss under the stated site, forecast package and scoring protocol.

A capacity-matched control strengthens this information assignment. With 20-dimensional weather summaries compared with timestamp-matched noise, the state-score gain remains 24.95–28.92% for NP025 and 13.97–16.25% for NP028 across the six leads; adding the same 20-dimensional weather summary to the 20-dimensional power arm improves the state score over the matched power-plus-noise arm by 5.47–11.93% for NP025 and 6.81–9.47% for NP028. The result identifies a weather-signal contribution beyond feature dimension alone, while the projection and timing controls are retained in Supplementary S18.

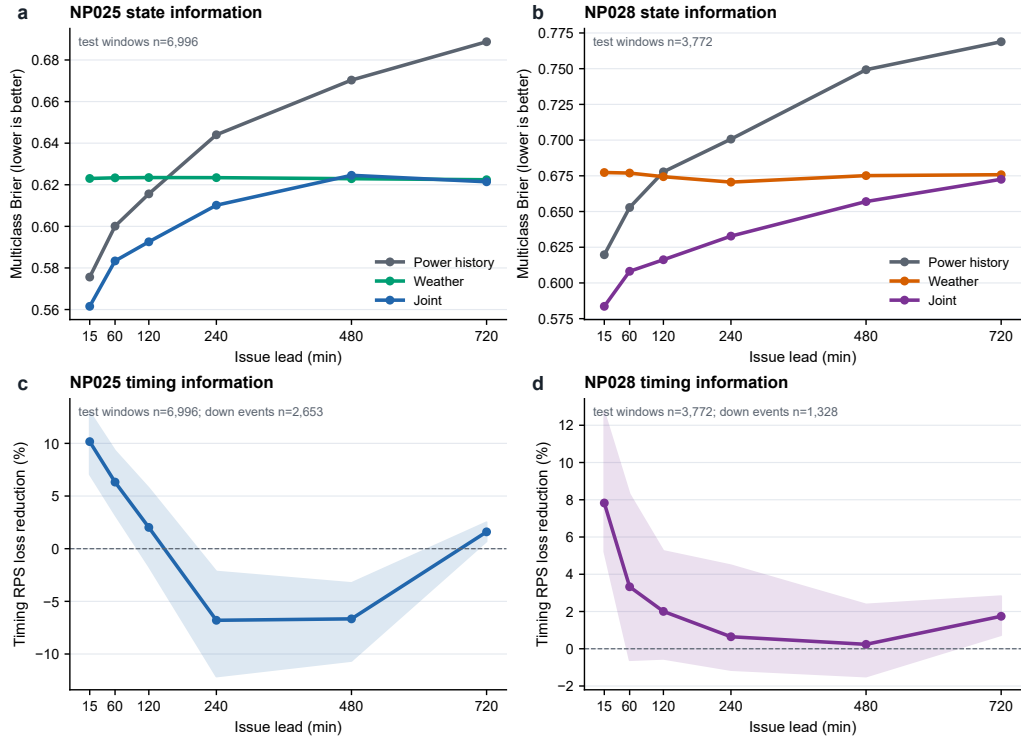


Figure 4: Absolute state scores and target-conditional information increments for NP025 and NP028. Lines show absolute test scores; ribbons show paired seven-day block bootstrap intervals on the matched Suining windows.

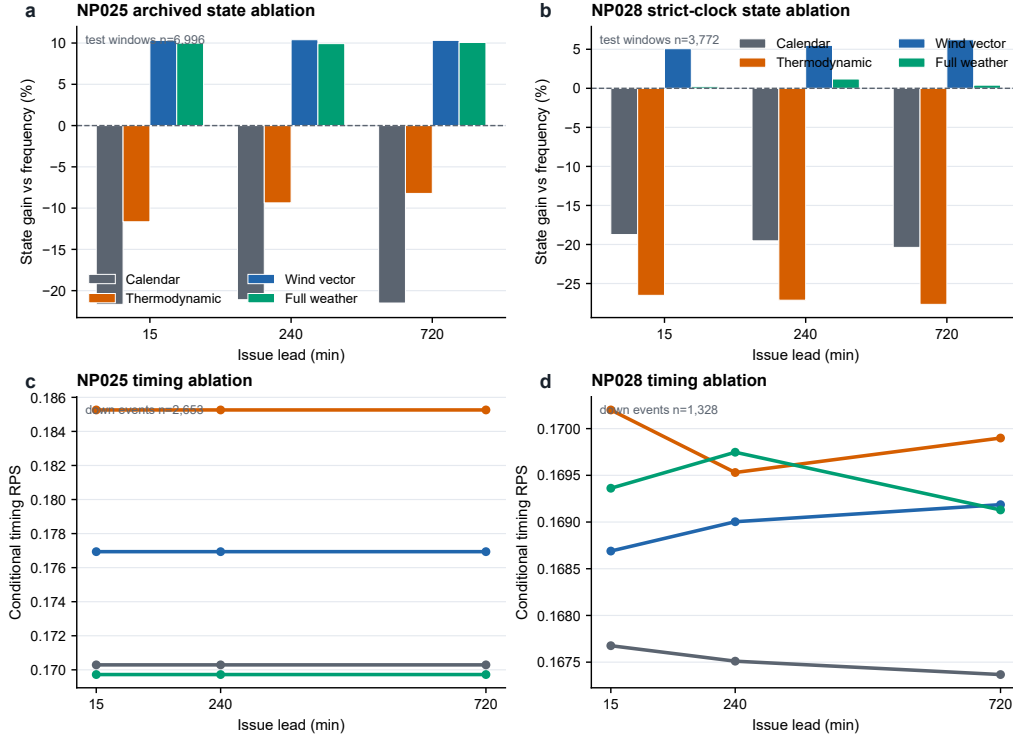


Figure 5: Weather-variable ablation for state and timing information. Bars compare calendar, thermodynamic, wind-vector and full-weather arms at 15, 240 and 720 min under the NP025/NP028 tree budgets.

3.3. The target-specific pattern persists under a fixed pre-issue weather clock

The strict-clock validation tests whether the information division persists within the NP028 Suining support window when the weather package is fixed before the valid target time (Fig. 4). The Suining previous_day1 package uses 100-m u/v wind, surface pressure and 2-m temperature, with complete weather support from 2024-02-16 and 3,772 test windows per lead. Every field is fixed 24 h before valid time. Relative to the power-history arm, adding this package improves the 12-h four-class Brier score by 12.11% (8.41–18.27%); relative to the weather arm, adding power history improves 15-min downward timing RPS by 7.83% (5.21–12.72%) and median-MAE by 7.40% (4.14–13.01%). Within this support window, the fixed clock preserves the target-specific division observed in the matched decomposition.

Under this fixed issue clock, the two information sources retain the same directional division of labour within the Suining support window. The routing test

reported next asks whether assigning them to separate heads improves the product itself.

3.4. Target-aligned routing improves intermediate-lead timing while preserving state risk

The routing experiment converts the decomposition into a process product (Fig. 6). NP026 reads only frozen NP025 predictions: the joint state head is retained, and the power-history timing head is assigned to downward first passage. The all-lead route landscape is shown in Fig. 6 and the monthly support audit in Fig. 7. In the archived package, the route improves timing RPS by 8.70% (4.96–12.84%) at 240 min and 8.65% (5.55–12.20%) at 480 min; the corresponding absolute conditional RPS changes from 0.183 to 0.167 and from 0.182 to 0.167. Median arrival-time MAE improves by 8.01% (5.10–10.97%) and 8.79% (5.00–12.94%). The state probabilities are identical by construction and by replay. A frozen UTC-month audit retains positive RPS and MAE increments at 240 and 480 min in June, July and August 2024.

The strict-clock routing replay retains positive RPS point estimates at all six leads, with the clearest interval support at 60 and 120 min. In the July–August audit, RPS and median-MAE point estimates are positive at 15, 60 and 120 min in both months; the longer-lead signs vary by month. The no-crossing-inclusive NP036 diagnostic adds a coherent product check: against the constructed joint distribution, the route improves RPS by 1.08% and 1.16% at 240 and 480 min in NP025, and by 0.47%, 0.76% and 0.62% at 240, 480 and 720 min in NP028. Against a directly refit 17-bin joint model, the strict-clock route improves by 5.97%, 11.20% and 10.84% at those three leads, while the archived-package comparison is mixed. The all-lead route summary gives the same pattern without selecting a local endpoint. Against the constructed joint product, the equal-weight six-lead conditional-RPS gain is 2.79% (1.01–4.89%) for NP025 and 1.45% (0.76–1.98%) for NP028; the corresponding median-MAE gains are 2.55% (0.79–4.29%) and 1.71% (0.60–2.46%). Against the directly refit 17-bin joint model, the strict NP028 route improves the equal-weight score by 5.82% (1.01–11.59%), whereas the archived package is –0.76% (–1.54 to 0.19%). These global estimates make the route a protocol-conditioned timing allocation result rather than an endpoint-selected comparison.

The same frozen-probability interface has a measurable decision value. Using validation-selected alert thresholds and a false-negative/false-positive cost ratio of 5, the all-lead full-rolling state product reduces test cost by 36.71–41.98% across the six leads, with the 12-h reduction equal to 41.98% (32.62–48.65%). At the

two leads where the earlier NP023 fusion candidates exist, NP019/NP020 reduce cost relative to NP014 by 3.40% at 12 h (−1.50 to 7.95%); this is a study-defined forecast-to-decision sensitivity rather than a market-cost claim (Fig. 8).

The route therefore supports a lead- and protocol-conditioned timing allocation rule while leaving the joint state-risk probabilities unchanged.

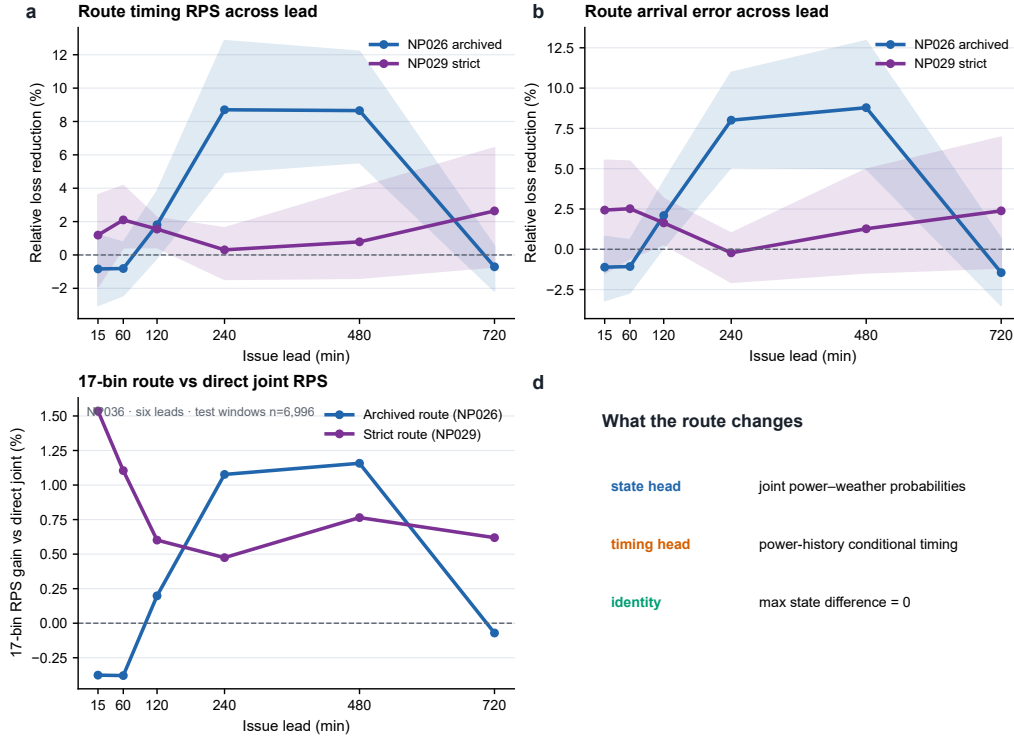


Figure 6: Target-aligned routing across all issue leads. Ribbons are paired seven-day block bootstrap intervals; archived NP026 uses 6,996 windows and strict NP029 uses 3,772 windows. The state identity panel reports the maximum absolute state-probability difference.

4. Discussion

4.1. A process view of predictability

The results support a process view of wind-power predictability under the tested protocols. Event-state risk and first-passage timing are distinct forecast objects with different information horizons. The matched information arms assign broad state-risk information to issued weather and near-term arrival information to power history, while range and path representations add amplitude and order

information as supporting objects. This view makes the information-to-target relationship measurable.

4.2. Information use should follow the target

The experiments support a target-aligned information rule under the matched Suining protocols. Within the fixed feature contracts, the archived weather trajectory is the arm associated with lower state-risk loss, while power history is the arm associated with lower first-passage timing loss; the strict previous_day1 package reproduces the state-risk increment within the NP028 Suining support window. Direct range modeling adds amplitude uncertainty information, and path assignment adds event-order information. The target-aligned route assigns the state and timing channels to the corresponding heads and improves intermediate-lead timing. The resulting design gives each information source a measurable process target and a stated issue-time contract.

4.3. Relation to the two existing studies

The published GNSS study established that an additional atmospheric source can enter a short-horizon power forecast³⁹. The present study extends that published forecast contribution by assigning information to event-state and first-passage objects and by turning the assignment into a target-aligned route.

4.4. Limitations and scope

The main information-arm study is centered on one Pizhou site and an explored calendar. NP022 adds an independent La Haute Borne site and a Suining process-object replication, while NP025 and NP028 add matched information decompositions at Suining. The archived NP025 weather trajectory is a retrospective information arm because its historical API archive does not expose a row-level model release timestamp; the strict issue-time claim is carried by NP028 within its Suining support window, where previous_day1 values are fixed 24 h before valid time. ERA5 and future-wind products provide retrospective information-gap diagnostics; causal atmospheric attribution requires a separate experimental design. NP025 uses 6,996 test windows, whereas NP028 uses 3,772 windows with the reduced previous_day1 variable set and complete weather support from 2024-02-16. The route reuses frozen heads and reports marginal state and timing products. A coherent joint path-time product is a defined extension of this framework. Proper-score improvement measures forecast loss; reliability and operational threshold performance are separate product properties. For example, representative 12-h joint downward-arrival reliability gaps reach 0.185 in NP025 and 0.290 in NP028

in Supplementary S12, so the state product is interpreted as a forecast-probability interface rather than a calibrated risk certificate. The complete lead-by-lead and monthly matrices, Pizhou core protocols, missingness audit and decision-cost sensitivity are retained in the Supplementary materials.

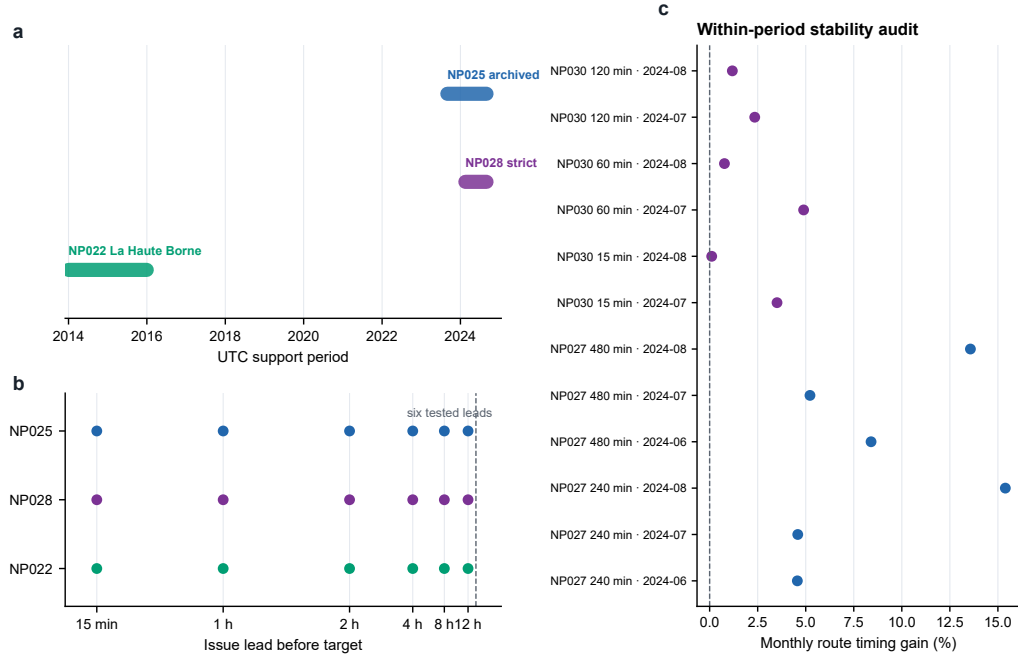


Figure 7: Support windows, issue-time clocks and within-period stability. The timeline marks the data and information contracts; the right panel reports frozen monthly point estimates from NP027/NP030.

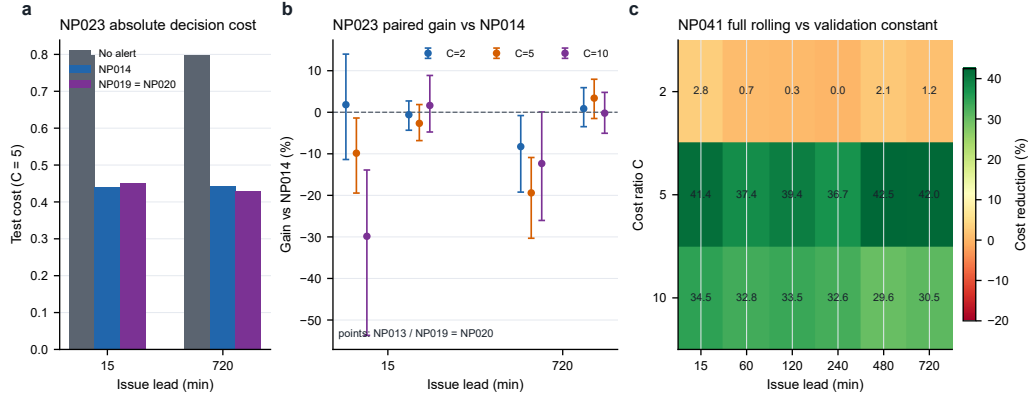


Figure 8: Forecast-to-decision sensitivity using frozen probabilities. Panel a shows all-lead full-rolling cost reduction versus no alert; panel b retains the NP023 fusion comparison at its available leads; panel c gives the all-lead cost-ratio surface.

5. Conclusion

Wind-power events contain forecast objects with different information horizons. In matched Suining experiments, the weather arm is associated with lower state-risk loss relative to the frequency reference, power history is associated with lower first-passage timing loss, and the joint arm retains complementary state information at longer leads. A fixed 24-h previous_day1 clock preserves the state-risk increment within the NP028 Suining support window. Target-aligned routing improves timing at documented leads—240 and 480 min in the archived package and 60 and 120 min in the strict-clock package—while preserving the joint state-risk probabilities; the archived direct-joint comparison remains mixed. These results support target-aligned information use for the tested wind-power process-forecasting protocol.

References

- [1] Y. Xie, C. Li, M. Li, F. Liu, M. Taukenova, An overview of deterministic and probabilistic forecasting methods of wind energy, *iScience* 26 (1) (2023) 105804. [doi:10.1016/j.isci.2022.105804](https://doi.org/10.1016/j.isci.2022.105804).
- [2] A. M. Foley, P. G. Leahy, A. Marvuglia, E. J. McKeogh, Current methods and advances in forecasting of wind power generation, *Renewable Energy* 37 (1) (2012) 1–8. [doi:10.1016/j.renene.2011.05.033](https://doi.org/10.1016/j.renene.2011.05.033).

- [3] J. Jung, R. P. Broadwater, Current status and future advances for wind speed and power forecasting, *Renewable and Sustainable Energy Reviews* 31 (2014) 762–777. doi:[10.1016/j.rser.2013.12.054](https://doi.org/10.1016/j.rser.2013.12.054).
- [4] T. Gneiting, M. Katzfuss, Probabilistic forecasting, *Annual Review of Statistics and Its Application* 1 (1) (2014) 125–151. doi:[10.1146/annurev-statistics-062713-085831](https://doi.org/10.1146/annurev-statistics-062713-085831).
- [5] C. Sweeney, R. J. Bessa, J. Browell, P. Pinson, The future of forecasting for renewable energy, *WIREs Energy and Environment* 9 (2) (2020) e365. doi:[10.1002/wene.365](https://doi.org/10.1002/wene.365).
- [6] C. Gallego-Castillo, A. Cuerva-Tejero, O. Lopez-Garcia, A review on the recent history of wind power ramp forecasting, *Renewable and Sustainable Energy Reviews* 52 (2015) 1148–1157. doi:[10.1016/j.rser.2015.07.154](https://doi.org/10.1016/j.rser.2015.07.154).
- [7] A. Bossavy, R. Girard, G. Kariniotakis, Forecasting ramps of wind power production with numerical weather prediction ensembles, *Wind Energy* 16 (1) (2013) 51–63. doi:[10.1002/we.526](https://doi.org/10.1002/we.526).
- [8] E. Ahn, J. Hur, A practical metric to evaluate the ramp events of wind generating resources to enhance the security of smart energy systems, *Energies* 15 (7) (2022) 2676. doi:[10.3390/en15072676](https://doi.org/10.3390/en15072676).
- [9] J. Fu, Y. Ni, Y. Ma, J. Zhao, Q. Yang, S. Xu, X. Zhang, Y. Liu, A visualization-based ramp event detection model for wind power generation, *Energies* 16 (3) (2023) 1166. doi:[10.3390/en16031166](https://doi.org/10.3390/en16031166).
- [10] A. Couto, P. Costa, L. Rodrigues, V. V. Lopes, A. Estanqueiro, Impact of weather regimes on the wind power ramp forecast in portugal, *IEEE Transactions on Sustainable Energy* 6 (3) (2015) 934–942. doi:[10.1109/tste.2014.2334062](https://doi.org/10.1109/tste.2014.2334062).
- [11] V. Q. Doan, H. Kusaka, M. Matsueda, R. Ikeda, Application of mesoscale ensemble forecast method for prediction of wind speed ramps, *Wind Energy* 22 (4) (2019) 499–508. doi:[10.1002/we.2302](https://doi.org/10.1002/we.2302).
- [12] M. Okada, K. Yamaguchi, R. Kodama, N. Ogasawara, H. Kato, V. Q. Doan, N. N. Ishizaki, H. Kusaka, Development of a wind power ramp forecasting system via meteorological pattern analysis, *Wind Energy* 25 (11) (2022) 1900–1916. doi:[10.1002/we.2774](https://doi.org/10.1002/we.2774).

- [13] M. Lochmann, H. Kalesse-Los, M. Schäfer, I. Heinrich, R. Leinweber, Analysing wind power ramp events and improving very short-term wind power predictions by including wind speed observations, *Wind Energy* 26 (6) (2023) 573–588. [doi:10.1002/we.2816](https://doi.org/10.1002/we.2816).
- [14] J. C. Lee, C. Draxl, L. K. Berg, Evaluating wind speed and power forecasts for wind energy applications using an open-source and systematic validation framework, *Renewable Energy* 200 (2022) 457–475. [doi:10.1016/j.renene.2022.09.111](https://doi.org/10.1016/j.renene.2022.09.111).
- [15] L. Bianco, R. Mendeke, J. Lindblom, I. V. Djalalova, D. D. Turner, J. M. Wilczak, Evaluating the ability of the operational high resolution rapid refresh model version 3 (hrrrv3) and version 4 (hrrrv4) to forecast wind ramp events in the us great plains, *Wind Energy Science* 10 (10) (2025) 2117–2136. [doi:10.5194/wes-10-2117-2025](https://doi.org/10.5194/wes-10-2117-2025).
- [16] R. P. Worsnop, M. Scheuerer, T. M. Hamill, J. K. Lundquist, Generating wind power scenarios for probabilistic ramp event prediction using multivariate statistical post-processing, *Wind Energy Science* 3 (1) (2018) 371–393. [doi:10.5194/wes-3-371-2018](https://doi.org/10.5194/wes-3-371-2018).
- [17] T. Muschinski, M. N. Lang, G. J. Mayr, J. W. Messner, A. Zeileis, T. Simon, Predicting power ramps from joint distributions of future wind speeds, *Wind Energy Science* 7 (6) (2022) 2393–2405. [doi:10.5194/wes-7-2393-2022](https://doi.org/10.5194/wes-7-2393-2022).
- [18] R. Schefzik, T. L. Thorarinsdottir, T. Gneiting, Uncertainty quantification in complex simulation models using ensemble copula coupling, *Statistical Science* 28 (4) (2013) 616–640. [doi:10.1214/13-sts443](https://doi.org/10.1214/13-sts443).
- [19] J. Tastu, P. Pinson, P.-J. Trombe, H. Madsen, Probabilistic forecasts of wind power generation accounting for geographically dispersed information, *IEEE Transactions on Smart Grid* 5 (1) (2014) 480–489. [doi:10.1109/tsg.2013.2277585](https://doi.org/10.1109/tsg.2013.2277585).
- [20] A. E. Raftery, T. Gneiting, F. Balabdaoui, M. Polakowski, Using bayesian model averaging to calibrate forecast ensembles, *Monthly Weather Review* 133 (5) (2005) 1155–1174. [doi:10.1175/mwr2906.1](https://doi.org/10.1175/mwr2906.1).

- [21] T. Gneiting, A. E. Raftery, Strictly proper scoring rules, prediction, and estimation, *Journal of the American Statistical Association* 102 (477) (2007) 359–378. [doi:10.1198/016214506000001437](https://doi.org/10.1198/016214506000001437).
- [22] T. Gneiting, F. Balabdaoui, A. E. Raftery, Probabilistic forecasts, calibration and sharpness, *Journal of the Royal Statistical Society Series B: Statistical Methodology* 69 (2) (2007) 243–268. [doi:10.1111/j.1467-9868.2007.00587.x](https://doi.org/10.1111/j.1467-9868.2007.00587.x).
- [23] H. Hersbach, Decomposition of the continuous ranked probability score for ensemble prediction systems, *Weather and Forecasting* 15 (5) (2000) 559–570. [doi:10.1175/1520-0434\(2000\)015<0559:DOTCRP>2.0.CO;2](https://doi.org/10.1175/1520-0434(2000)015<0559:DOTCRP>2.0.CO;2).
- [24] A. Jordan, F. Krüger, S. Lerch, Evaluating probabilistic forecasts with scoringrules, *Journal of Statistical Software* 90 (12) (2019) 1–37. [doi:10.18637/jss.v090.i12](https://doi.org/10.18637/jss.v090.i12).
- [25] R. Taggart, Evaluation of point forecasts for extreme events using consistent scoring functions, *Quarterly Journal of the Royal Meteorological Society* 148 (742) (2021) 306–320. [doi:10.1002/qj.4206](https://doi.org/10.1002/qj.4206).
- [26] T. Gneiting, R. Ranjan, Comparing density forecasts using threshold- and quantile-weighted scoring rules, *Journal of Business & Economic Statistics* 29 (3) (2011) 411–422. [doi:10.1198/jbes.2010.08110](https://doi.org/10.1198/jbes.2010.08110).
- [27] C. Liu, X. Zhang, S. Mei, Z. Zhen, M. Jia, Z. Li, H. Tang, Numerical weather prediction enhanced wind power forecasting: Rank ensemble and probabilistic fluctuation awareness, *Applied Energy* 313 (2022) 118769. [doi:10.1016/j.apenergy.2022.118769](https://doi.org/10.1016/j.apenergy.2022.118769).
- [28] C. Liu, X. Zhang, S. Mei, Q. Zhou, H. Fan, Series-wise attention network for wind power forecasting considering temporal lag of numerical weather prediction, *Applied Energy* 336 (2023) 120815. [doi:10.1016/j.apenergy.2023.120815](https://doi.org/10.1016/j.apenergy.2023.120815).
- [29] Y. Hu, H. Liu, S. Wu, Y. Zhao, Z. Wang, X. Liu, Temporal collaborative attention for wind power forecasting, *Applied Energy* 357 (2024) 122502. [doi:10.1016/j.apenergy.2023.122502](https://doi.org/10.1016/j.apenergy.2023.122502).

- [30] Y. Liu, J. Wang, L. Liu, Physics-informed reinforcement learning for probabilistic wind power forecasting under extreme events, *Applied Energy* 376 (2024) 124068. [doi:10.1016/j.apenergy.2024.124068](https://doi.org/10.1016/j.apenergy.2024.124068).
- [31] L. de Azevedo Takara, A. C. Teixeira, H. Yazdanpanah, V. C. Mariani, L. dos Santos Coelho, Optimizing multi-step wind power forecasting: Integrating advanced deep neural networks with stacking-based probabilistic learning, *Applied Energy* 369 (2024) 123487. [doi:10.1016/j.apenergy.2024.123487](https://doi.org/10.1016/j.apenergy.2024.123487).
- [32] D. Nohara, M. Ohba, T. Watanabe, S. Kadokura, Probabilistic wind power prediction based on ensemble weather forecasting, *IFAC-PapersOnLine* 53 (2) (2020) 12151–12156. [doi:10.1016/j.ifacol.2020.12.983](https://doi.org/10.1016/j.ifacol.2020.12.983).
- [33] D. Salinas, V. Flunkert, J. Gasthaus, T. Januschowski, Deepar: Probabilistic forecasting with autoregressive recurrent networks, *International Journal of Forecasting* 36 (3) (2020) 1181–1191. [doi:10.1016/j.ijforecast.2019.07.001](https://doi.org/10.1016/j.ijforecast.2019.07.001).
- [34] J. Gasthaus, K. Benidis, Y. Wang, S. S. Rangapuram, D. Salinas, V. Flunkert, T. Januschowski, [Probabilistic forecasting with spline quantile function rnns](https://proceedings.mlr.press/v89/gasthaus19a.html), in: *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, Vol. 89, PMLR, 2019, pp. 1901–1910. URL <https://proceedings.mlr.press/v89/gasthaus19a.html>
- [35] C. Jin, Y. Yang, C. Han, T. Lei, C. Li, B. Lu, Evaluation of forecasted wind speed at turbine hub height and wind ramps by five nwp models with observations from 262 wind farms over china, *Meteorological Applications* 31 (6) (2024) e70007. [doi:10.1002/met.70007](https://doi.org/10.1002/met.70007).
- [36] B. R. Cheneka, S. J. Watson, S. Basu, Quantifying the impacts of synoptic weather patterns on north sea wind power production and ramp events under a changing climate, *Energy and Climate Change* 4 (2023) 100113. [doi:10.1016/j.egycc.2023.100113](https://doi.org/10.1016/j.egycc.2023.100113).
- [37] J. Olauson, Era5: The new champion of wind power modelling?, *Renewable Energy* 126 (2018) 322–331. [doi:10.1016/j.renene.2018.03.056](https://doi.org/10.1016/j.renene.2018.03.056).
- [38] J. Yan, C. Möhrle, T. Göçmen, M. Kelly, A. Wessel, G. Giebel, Uncovering wind power forecasting uncertainty sources and their propagation through

the whole modelling chain, *Renewable and Sustainable Energy Reviews* 165 (2022) 112519. [doi:10.1016/j.rser.2022.112519](https://doi.org/10.1016/j.rser.2022.112519).

- [39] P. Gong, C. Yuan, Ground-based gnss atmospheric remote sensing for ultra-short-term wind-power forecasting: A direction-proxy-guided graph-residual approach, *Remote Sensing* 18 (18) (2026) 3095. [doi:10.3390/rs18183095](https://doi.org/10.3390/rs18183095).
- [40] M. Korpaas, A. T. Holen, R. Hildrum, Operation and sizing of energy storage for wind power plants in a market system, *International Journal of Electrical Power & Energy Systems* 25 (8) (2003) 599–606. [doi:10.1016/s0142-0615\(03\)00016-4](https://doi.org/10.1016/s0142-0615(03)00016-4).
- [41] T. Hong, S. Fan, Probabilistic electric load forecasting: A tutorial review, *International Journal of Forecasting* 32 (3) (2016) 914–938. [doi:10.1016/j.ijforecast.2015.11.011](https://doi.org/10.1016/j.ijforecast.2015.11.011).
- [42] L. J. Tashman, Out-of-sample tests of forecasting accuracy: an analysis and review, *International Journal of Forecasting* 16 (4) (2000) 437–450. [doi:10.1016/s0169-2070\(00\)00065-0](https://doi.org/10.1016/s0169-2070(00)00065-0).
- [43] V. Cerqueira, L. Torgo, I. Mozetič, Evaluating time series forecasting models: an empirical study on performance estimation methods, *Machine Learning* 109 (11) (2020) 1997–2028. [doi:10.1007/s10994-020-05910-7](https://doi.org/10.1007/s10994-020-05910-7).
- [44] J. Perr-Sauer, M. Optis, J. Fields, N. Bodini, J. Lee, A. Todd, E. Simley, R. Hammond, C. Phillips, M. Lunacek, T. Kemper, L. Williams, A. Craig, N. Agarwal, S. Sheng, J. Meissner, Openoa: An open-source codebase for operational analysis of wind farms, *Journal of Open Source Software* 6 (58) (2021) 2171. [doi:10.21105/joss.02171](https://doi.org/10.21105/joss.02171).
- [45] H. Hersbach, B. Bell, P. Berrisford, S. Hirahara, A. Horányi, J. Muñoz-Sabater, J. Nicolas, C. Peubey, R. Radu, D. Schepers, A. Simmons, C. Soci, S. Abdalla, X. Abellan, G. Balsamo, P. Bechtold, G. Biavati, J. Bidlot, M. Bonavita, G. D. Chiara, P. Dahlgren, D. Dee, M. Diamantakis, R. Dragani, J. Flemming, R. Forbes, M. Fuentes, A. Geer, L. Haimberger, S. Healy, R. J. Hogan, E. Hólm, M. Janisková, S. Keeley, P. Laloyaux, P. Lopez, C. Lupu, G. Radnoti, P. de Rosnay, I. Rozum, F. Vamborg, S. Villaume, J.-N. Thépaut, The era5 global reanalysis, *Quarterly Journal of the Royal Meteorological Society* 146 (730) (2020) 1999–2049. [doi:10.1002/qj.3803](https://doi.org/10.1002/qj.3803).

- [46] N. Meinshausen, [Quantile regression forests](https://www.jmlr.org/papers/volume7/meinshausen06a/meinshausen06a.html), Journal of Machine Learning Research 7 (35) (2006) 983–999.
URL <https://www.jmlr.org/papers/volume7/meinshausen06a/meinshausen06a.html>

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CRedit authorship contribution statement

Peiyan Gong: Conceptualization, methodology, software, formal analysis, visualization, writing – original draft. Chaoxia Yuan: Supervision, project administration, funding acquisition, writing – review and editing. Jie Tian: Validation, data curation, writing – review and editing. Yanxi Shi: Validation, data curation, writing – review and editing. All authors approved the final manuscript.

Declaration of competing interest

The authors declare no competing interests.

Data and code availability

Code will be released publicly. Public-source wind-farm datasets will be released or linked with their source records and licenses. Private wind-farm SCADA, coordinates and operational-status records will not be released or redistributed.